HealthTrack Alert Configuration: Thresholds, Rules, and Escalation

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HealthTrack checks each reading as it arrives and runs trend rules every 15 minutes. Tables 1 to 3 give the default rules, the escalation policies, and how each setting is changed.

Thresholds and Rules

Each rule has a program default, but the value applied to a reading is resolved per patient in a fixed order: an active patient override, then the patient's current exercise prescription, then the default. Each alert records which source applied. Rules that need two consecutive readings filter out single motion artifacts, and readings flagged as suspect never alert.

Table 1

Default Alert Rules

Rule	Condition (default)	Personalized by	Readings	Severity	Suppression
HR_ABOVE_CEILING	Exercise heart rate > 150 bpm	Prescription	2 in 30 min	Critical	Never
SPO2_LOW	SpO2 < 90%, any context	Prescription	2 in 30 min	Critical	Never
EXERCISE_SBP_HIGH	Exercise systolic > 200 mmHg	Prescription	1	High	30 min
RESTING_SBP_LOW	Resting systolic < 90 mmHg	Override	1	High	4 h
RESTING_HR_LOW	Resting heart rate < 40 bpm	Override	2 in 60 min	High	4 h
MISSED_SESSIONS	2 prescribed sessions missed in a row	Override	Trend	High	24 h
RESTING_SBP_HIGH	Resting systolic 160 mmHg or more	Override	1	Moderate	12 h
RESTING_HR_HIGH	Resting heart rate > 100 bpm	Override	2 in 60 min	Moderate	12 h
RESTING_HR_RISE	7-day resting HR > baseline + 8 bpm	Override	Trend	Moderate	24 h
ZONE_MINUTES_DECLINE	Zone minutes down 25% week over week	Override	Trend	Moderate	24 h
STEPS_DECLINE	Daily steps down 30% from prior week	Override	Trend	Moderate	24 h
DATA_GAP	No readings for 48 hours	Override	Scheduled	Low	24 h

Note. Exercise rules use warmup, exercise, and cooldown readings; resting rules use rest readings only. Suppression is how long a repeat of the rule after a recent alert is logged without notifying.

Prioritization, Alert Fatigue, and Escalation

Alarm fatigue is a recognized patient safety hazard because clinicians who receive frequent low-value alarms begin to miss the ones that matter (The Joint Commission, 2013). HealthTrack applies four controls in order: a repeat of an active alert is folded into it, a non-critical rule that fired recently is logged as suppressed for its window, a lower-severity alert on a measurement with a more severe active alert is masked, and a patient with more than four non-critical alerts in an hour has email and SMS moved to the digest. Critical alerts skip every

control. Email and SMS carry only the severity, rule name, and a sign-in link, following the minimum necessary standard (U.S. Department of Health and Human Services, n.d.). An alert nobody acknowledges within its policy timeout moves up one level and notifies the next recipients (Table 2); acknowledging stops escalation, and the alert stays on the worklist until it is resolved with a note.

Table 2

Escalation Policies

Severity	First notified	Level 1	Level 2
Critical	Clinicians: dashboard, SMS, email	Supervising physician, 15 min	Whole care team, all channels, 30 min
High	Clinicians: dashboard, email	Supervising physician, 60 min	Whole care team, all channels, 120 min
Moderate	Clinicians: dashboard	Supervising physician, 4 h	None
Low	Clinicians: dashboard; email in digest	None	None

Note. Clinicians are the patient's primary and covering clinicians. Each level's timer starts when the previous level fired.

Setting and Modifying Configuration

Each setting is changed through one API resource, limited to the role accountable for it (Table 3). Every change needs a written reason, is audited, keeps the earlier version, and applies from the next reading or scheduled check. GET /patients/{id}/alert-thresholds shows each value in force and its source.

Table 3

How Alert Configuration Is Changed

Setting	API call	Who	Validation
Program rule: default, readings, window, severity, suppression, active	PATCH /alert-rules/{code}	Physician	Critical rules keep suppression at zero
Prescription: ceiling, target zone, SpO2 floor, exercise systolic, resting baseline	PUT /patients/{id}/prescription	Physician on care team	Resting < zone < ceiling; old version closed
Override of a customizable rule for one patient	POST or DELETE /patients/{id}/alert-thresholds	Care team clinician or physician	Reason required; not for prescription-driven rules
Escalation policy: timeout, role, levels, channels, digest	PUT /escalation-policies/{severity}	Physician or administrator	Critical within 30 min; digest for low only; dashboard always on

Note. Paths are under /api/v1. Job intervals, the digest hour, and the storm limit are environment settings.

References

The Joint Commission. (2013, April 8). Medical device alarm safety in hospitals. *Sentinel Event Alert*, (50). https://www.jointcommission.org/sea_issue_50

U.S. Department of Health and Human Services. (n.d.). *Minimum necessary requirement*. Retrieved November 10, 2026, from https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/minimum-necessary-requirement/index.html